

ATTENDANCE MANAGEMENT USING FACIAL RECOGNITION

*Mini-Project Report submitted to the SASTRA Deemed to be University
in partial fulfilment of the requirements
for the award of the degree of*

B. Tech. Electronics and Communication Engineering

Submitted by

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May 2026



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Bonafide Certificate

This is to certify that the report titled “**Attendance Management System Using Facial Recognition**” submitted in partial fulfilment of the requirements for the award of the degree of B. Tech. Electronics and Communication Engineering to the SASTRA Deemed to be University, is a bona-fide record of the work done by **Abarna Sivaprasad (Reg no: 127004002), Ragul V S (Reg no: 127004199), Rishivanth R R (Reg no: 127004215)** during the Sixth semester of the academic year 2025-26, in the **School of Electrical & Electronics Engineering**, under my supervision. This report has not formed the basis for the award of any degree, diploma, associateship, fellowship or other similar title to any candidate of any University

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Project *Viva voce* held on: 07/05/2026

Examiner 1 Examiner 2



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Declaration

We declare that the report titled “**Attendance Management System Using Facial Recognition**” submitted by us is an original work done by us under the guidance of **Dr. Rajamohan J, Sr. Assistant Professor, School of Electrical and Electronics Engineering, SASTRA Deemed to be University** during the sixth semester of the academic year 2025-26, in the **School of Electrical and Electronics Engineering**. The work is original and whenever We have used materials from other sources, we have given due credit and cited them in the text of the report. This report has not formed the basis for the award of any degree, diploma, associateship, fellowship or other similar title to any candidate of any University.



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ABSTRACT

The aim of this project was to design an automated attendance system based on face recognition to increase the accuracy rate and remove the need for manual operations. This involves image acquisition, feature extraction using the FaceNet algorithm, and comparison with the stored embeddings to confirm the identity of the individual.

My research involved the pre-processing and data analysis part of the project. This entailed collecting the datasets, detecting faces, and generating facial embeddings.

Specific Contribution:

Data Acquisition: Facial data was captured using the camera module in real time and arranged for use in the system.

Pre-processing Design: Various pre-processing methods were developed, such as face detection, cropping, resizing, and normalization to maintain consistency in the data inputs.

Face Detection Module: Face detection through the use of Haar Cascade algorithm was incorporated into the system.

Data Preparation: Data embeddings for each of the individuals registered in the system were generated and stored.

Specific Learning:

Acquired knowledge of methods for pre-processing images, including face detection, normalization, and preparation of features for deep learning algorithms.

Acquired experience with acquiring live data and constructing datasets for use in machine learning algorithms.

Technical Limitations & Challenges Faced:

Detection Accuracy: Variations in lighting and face orientation affected detection performance, requiring adjustments in preprocessing techniques.

Data Consistency: Maintaining uniform image quality across datasets was challenging and required

proper normalization and filtering methods.

Keywords: *Face Recognition, Data Preprocessing, Haar Cascade, Face Detection, Deep Learning, Image Processing*

A handwritten signature in blue ink, appearing to read 'Dr. J. Rajamohan', with a long horizontal stroke extending to the right.

Signature of the Guide
Name: Dr. J.Rajamohan

Student Reg No: 127004002
Name: Abarna Sivaprasad

ABSTRACT

The current research involves the creation of an automated attendance system that uses face recognition technology in order to increase its accuracy and automate all the process related to the identification of individuals.

In my case, my work involved designing and integrating the FaceNet model in addition to creating the system with the help of the recognition algorithm based on Euclidean distance.

Specific Contribution:

Architecture Design: Designed the complete system architecture, defining the workflow from image acquisition to attendance recording.

FaceNet Integration: Implemented the FaceNet model for generating facial embeddings and ensured accurate feature extraction.

Recognition Logic: Developed the identity verification mechanism using Euclidean distance and established threshold values for face matching.

System Optimization: Tuned system parameters to ensure stable real- time performance with high recognition accuracy and reduced processing delay.

Specific Learning:

Acquired thorough knowledge about face recognition systems based on deep learning and embeddings.

Acquired knowledge about designing end-to-end pipelines for image processing and

decision-making systems.

Technical Limitations & Challenges Faced:

Lighting Differences: The impact of lighting conditions on face recognition accuracy necessitated preprocessing modifications to normalize results.

Real-time Requirements: The need for low latency and accurate face recognition demanded optimal model performance and parameter tuning.

Keywords: *Face Recognition, FaceNet, Automated Attendance System, Euclidean Distance, Deep Learning, Real-Time System*



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ABSTRACT

This study involves the design of a face recognition-based automated attendance system that enhances precision through automation and removes any human error in the process. The system takes snapshots, detects the face, extracts features from the faces using FaceNet, and matches these features against known database features to confirm the identity of individuals. In my case, the main areas of concentration were on implementing this system as well as managing attendance.

Specific Contribution:

Embedded System Integration: Installed the entire system on the Raspberry Pi 4 board and successfully integrated the hardware and software components of the system.

Attendance Recording Module: Developed an attendance system using CSV that could record name, date, and time automatically.

Duplicate Attendance Prevention: Created a system to ensure that no duplicate attendance was recorded for a single user.

Testing and Verification: Conducted live testing of the system to check its output accuracy and performance.

Specific Learning:

Gained practical experience in deploying machine learning systems on embedded platforms like Raspberry Pi 4

Learned how to design efficient attendance management systems with real-time logging and data validation techniques.

Technical Limitations & Challenges Faced:

Hardware Constraints: Limited processing power of the embedded system required optimization for smooth real-time execution.

Data Consistency: Ensuring accurate attendance logging without duplicate entries required careful session management and validation logic.

Keywords: *Face Recognition, Attendance System, Embedded Systems, Raspberry Pi, CSV Logging, Real-Time Processing*



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NOTATION

This section defines the symbols and notations used throughout the report titled "Attendance management system using face recognition".

Symbol	Description
X_i	Feature value of detected face
Y_i	Feature value of stored face
D	Euclidean distance
N	Number of features

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CHAPTER 1

INTRODUCTION

Attendance management plays a vital role in educational institutions and organizational environments by ensuring proper record keeping, monitoring participation, and maintaining discipline. With the rapid advancement of digital technologies, there is an increasing demand for automated and intelligent systems that can handle attendance efficiently and accurately. In this context, computer vision and machine learning have enabled the development of biometric identification techniques, among which facial recognition has emerged as a reliable and non-intrusive solution, as it does not require physical contact or active user involvement. This project presents an automated attendance management system using facial recognition technology, designed for real-time deployment on embedded platforms. The system utilizes a camera module to continuously capture video frames and detect human faces using computer vision techniques. The detected faces are processed using a deep learning-based FaceNet model, which generates high-dimensional embeddings that uniquely represent each individual. These embeddings are compared using a distance-based metric to accurately identify individuals. To ensure robustness in real-world conditions, preprocessing techniques and a threshold-based matching mechanism are incorporated to handle variations in lighting, pose, and minor occlusions. Once a face is successfully recognized, attendance is automatically recorded along with the corresponding date and time, eliminating manual intervention. Additionally, duplicate entry prevention mechanisms ensure that each individual is marked present only once per session, thereby maintaining data integrity. The system is implemented on Raspberry Pi 4, making it a cost-effective, portable, and energy-efficient solution suitable for deployment in classrooms, laboratories, and organizational environments. Its modular architecture allows easy enrollment of new users without requiring retraining of the model, thereby enhancing scalability and usability. Overall, the proposed system improves accuracy, reduces administrative workload, enhances security, and provides a seamless and automated solution for modern attendance management. Furthermore, it offers potential for future integration with smart campus systems, cloud-based data management, and enterprise-level applications.

1.1 Problem Statement

Traditional attendance systems rely heavily on manual processes, making them time-consuming, error-prone, and difficult to manage at scale. They are susceptible to issues like proxy attendance and inaccurate record-keeping, while growing data volumes further complicate organization and reliability. Even semi-automated solutions such as RFID or QR codes still depend on user interaction and do not fully prevent misuse. Therefore, there is a need for a fully automated, efficient, and reliable attendance system that can accurately identify individuals without manual intervention, ensuring data integrity and ease of management.

1.2 Literature Survey

TITLE OF THE PAPER	YEAR OF PUBLICATION	AUTHOR
[1] FaceNet: A Unified Embedding for Face Recognition and Clustering	2015	Florian Schroff, Dmitry Kalenichenko, James Philbin
[2] Automated Attendance System using Face Recognition on Raspberry Pi	2020	S. Arul, R. Karthik, P. Mohan
[3] Real-Time Face Recognition System for Attendance using Deep Learning	2022	A. Sharma, V. Gupta
[4] Smart Attendance System using Face Recognition and IoT	2021	N. Prakash, M. Ramesh

CHAPTER 2

OBJECTIVE

The main objective of this project is to develop a reliable, fully automated attendance management system using facial recognition technology that operates efficiently in real-time environments. The system is designed to detect and recognize human faces from live video input using computer vision techniques and to utilize a deep learning-based model (FaceNet) for extracting distinctive facial features as embeddings for accurate identification. A distance-based comparison mechanism is implemented to ensure precise identity verification, while preprocessing techniques are incorporated to handle variations in lighting, facial pose, and minor occlusions, ensuring consistent performance. Upon successful recognition, the system automatically records attendance with the corresponding date and time, minimizing manual effort, and includes a duplicate prevention mechanism to ensure each individual is marked present only once per session. Additionally, the system is deployed on a raspberry Pi platform to provide a compact, cost-effective, and energy-efficient solution suitable for real-world applications, and is designed in a modular manner to allow easy addition of new users without requiring complete model retraining.

CHAPTER 3

EXPERIMENTAL WORK / METHODOLOGY

This chapter describes the complete working mechanism of the proposed facial recognition-based attendance system. The system is implemented as a real-time processing pipeline on a Raspberry Pi 4, where each stage performs a specific operation on the incoming video frames to achieve accurate identification and automated attendance recording.

The processing sequence consists of: (i) continuous frame acquisition, (ii) face detection and region extraction, (iii) preprocessing and normalization, (iv) feature extraction using FaceNet, (v) embedding comparison using a distance metric, and (vi) attendance logging with duplicate prevention. All operations are executed locally on the embedded platform to maintain real-time performance without external dependency.

3.1 Image Acquisition and Frame Handling

A Pi Camera module is interfaced with the Raspberry Pi 4 to capture live video. Frames are read sequentially at a controlled rate to ensure stable processing without frame drops.

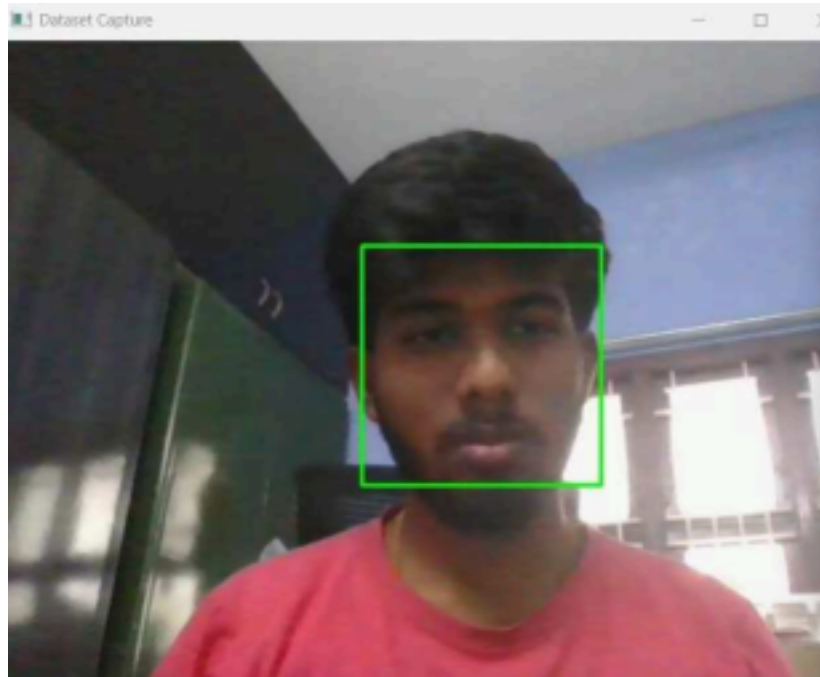
Each frame is buffered and passed immediately to the detection stage. The frame resolution is maintained at a moderate size to balance detection accuracy and processing speed. No frames are stored permanently at this stage; processing is performed in-memory to reduce latency.

3.2 Face Detection and Region Extraction

Face detection is performed on each frame using a Haar Cascade classifier. The classifier scans the image across multiple scales and identifies regions that satisfy predefined facial feature patterns.

For every detected face:

- A bounding box is generated indicating the face location.
- The region inside the bounding box is cropped from the original frame.



[Figure 3.1: Dataset Capture-bounding box]

If multiple faces are present, each detected region is processed independently in the same pipeline. If no face is detected in a frame, the system skips further processing and proceeds to the next frame.

3.3 Preprocessing and Normalization

Each cropped face region undergoes preprocessing to ensure uniform input to the recognition model:

Resizing: The face image is resized to the input dimension required by FaceNet.

Pixel Normalization: Pixel values are scaled to a consistent range to reduce the effect of illumination differences.

Histogram Equalization (if applied): Enhances contrast in low-light conditions.

Bounding Box Normalization: Ensures the face is centered and occupies a consistent area of the image. These steps reduce variability caused by lighting, scale, and minor pose differences, thereby improving recognition stability.

3.4 Data Augmentation

To improve the robustness of the face recognition system, data augmentation is applied during the dataset preparation stage. Instead of relying only on the original captured images, multiple variations of each face sample are generated. This helps the system learn to recognize individuals even when there are slight changes in appearance or environmental conditions.

In this project, each captured face image is transformed using a set of augmentation techniques. Horizontal flipping is used to simulate left–right variations in face orientation. Brightness adjustment is applied to represent different lighting conditions. Rotation is introduced to account for slight head tilts, and blurring is used to mimic minor camera or motion effects. These transformations ensure that the dataset contains diverse representations of the same individual.

The augmented images are stored along with the original images in the dataset directory, as shown in the figure below. Each variation is labeled systematically (such as flip, bright, blur, and rot), making it easier to organize and process during the embedding generation stage. By increasing the diversity of the dataset, the system becomes more capable of handling real-world variations in lighting, pose, and image quality.

Overall, data augmentation enhances the reliability of the recognition process by allowing the model to generalize better across different conditions, thereby improving the consistency of attendance marking.



[Figure 3.2: Augmented Dataset]

3.5 Feature Extraction using FaceNet

- The preprocessed face image is provided as input to the FaceNet model. FaceNet is a deep neural network that converts the image into a 128-dimensional embedding vector.
- Each embedding represents the identity-specific features of a face.
- The mapping is designed such that embeddings of the same person are close in the vector space, while those of different individuals are far apart.
- The generated embedding is a compact numerical representation and is used for comparison in the next stage.

3.6 Reference Database and Embedding Storage

Before real-time operation, each authorized user is enrolled into the system:

- Multiple face samples of the individual are captured.
- FaceNet is used to generate embeddings for these samples.
- The embeddings are stored in a local database along with the corresponding identity label.

During recognition, this database serves as the reference set for matching.

3.7 Face Recognition using Distance-Based Matching

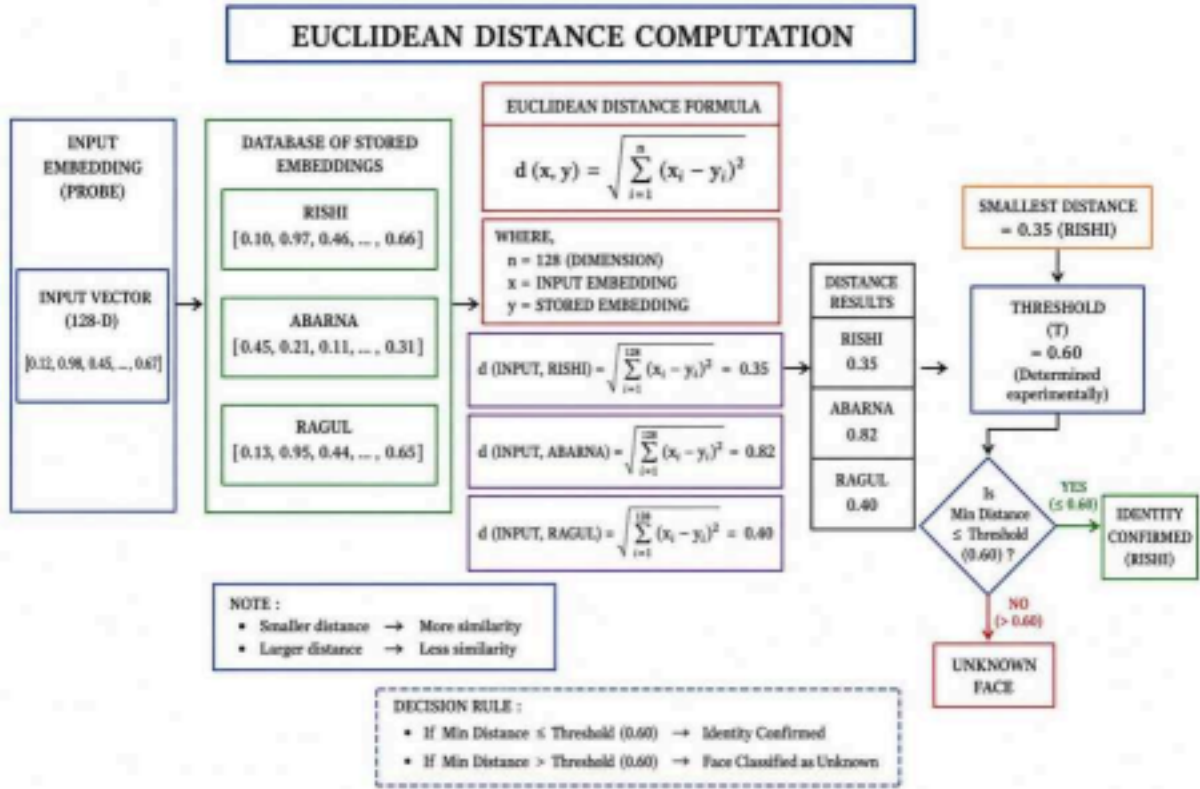
For each input embedding, the system computes its similarity with stored embeddings:

Euclidean distance is calculated between the input embedding and each reference embedding.

The smallest distance value is selected as the best match.

Decision rule:

- If the minimum distance $\leq$ predefined threshold $\rightarrow$ Identity confirmed •
- If the minimum distance $>$ threshold $\rightarrow$ Face classified as unknown



[Figure 3.3:Euclidean Distance Computation]

The threshold value is determined experimentally to ensure a balance between false acceptance and false rejection.

3.8 Decision Handling and Unknown Face Rejection

If no matching embedding satisfies the threshold condition:

The detected face is labeled as unknown.

No attendance entry is generated.

This prevents incorrect marking of attendance for unregistered individuals.

3.9 Duplicate Entry Prevention Mechanism

To maintain accurate attendance records, the system includes a session-based duplicate check:

When a person is recognized, the system verifies whether their attendance has already been recorded in the current session.

If an entry exists, no new record is created.

If no entry exists, the system proceeds to log attendance.

This ensures that each individual is recorded only once per session.

3.10 Attendance Recording and Data Storage

Once recognition is confirmed and duplicate check is cleared:

The system retrieves the identity label of the individual.

The current date and time are obtained from the system clock.

A new entry is appended to a CSV file in the format:

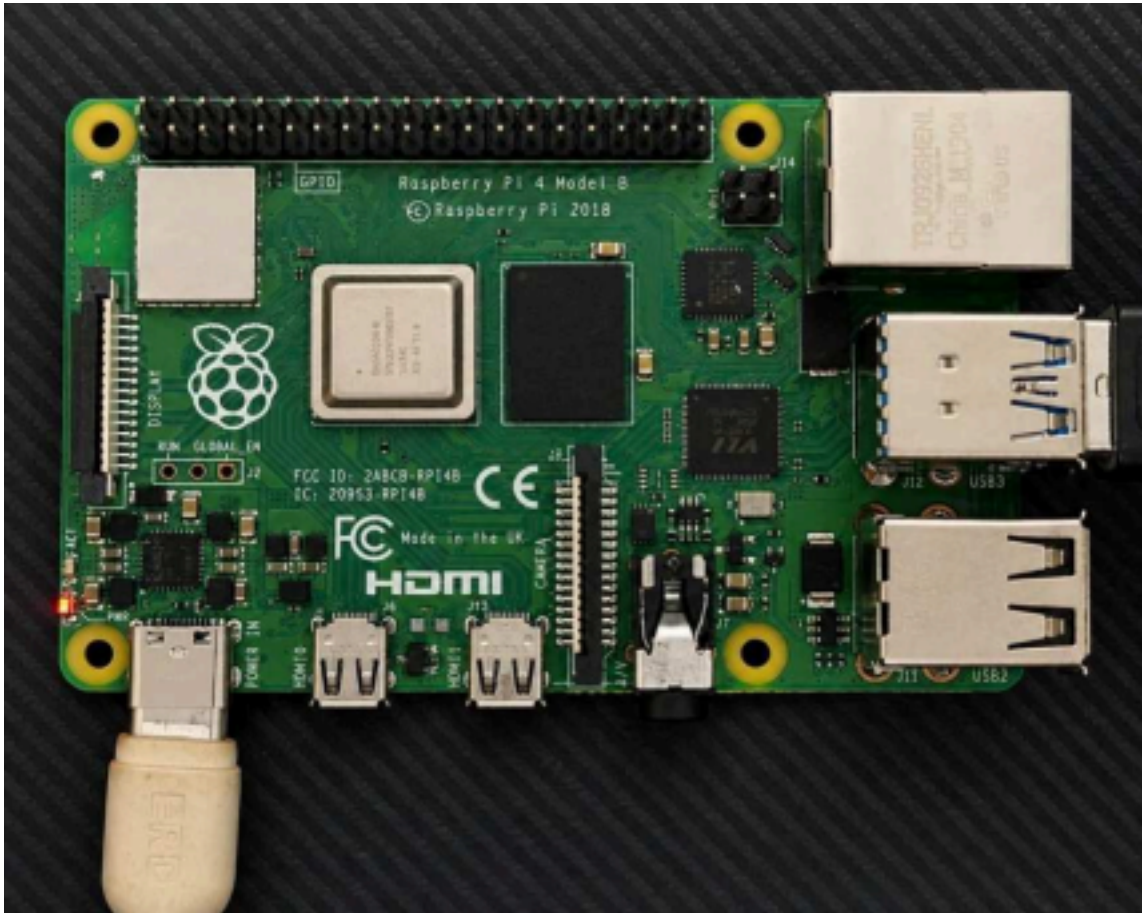
[Name, Date, Time]

The CSV format is used for its simplicity and compatibility with external analysis tools.

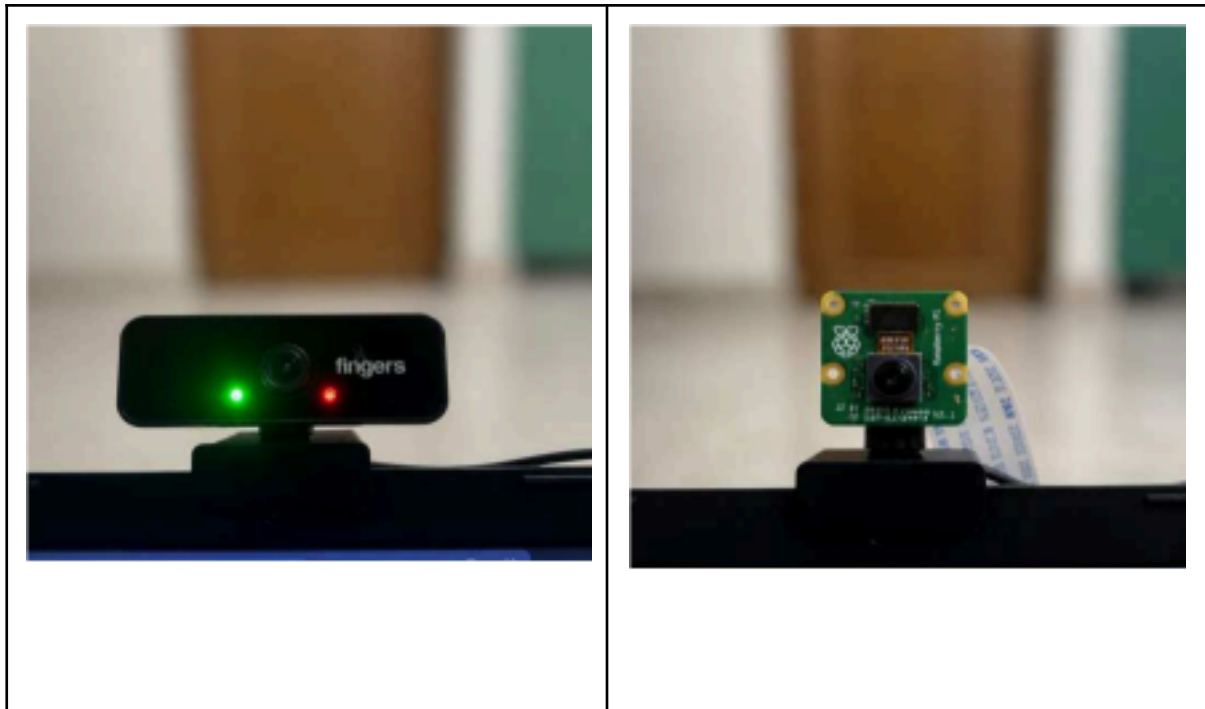
3.11 Hardware Implementation

The entire system is deployed on a Raspberry Pi 4, which acts as the central processing unit. The Pi Camera module provides continuous video input, and all computations including detection, feature extraction, and matching are performed locally.

The system is optimized to achieve real-time performance by controlling frame rate, minimizing memory usage, and avoiding redundant computations. No external server or cloud processing is required, ensuring a standalone and portable solution.



[Figure 3.4: Raspberry Pi 4]



[Figure 3.5:Hi-Res Webcam]

[Figure 3.6: Raspberry Pi V2 Cam]

3.12 System Execution Flow

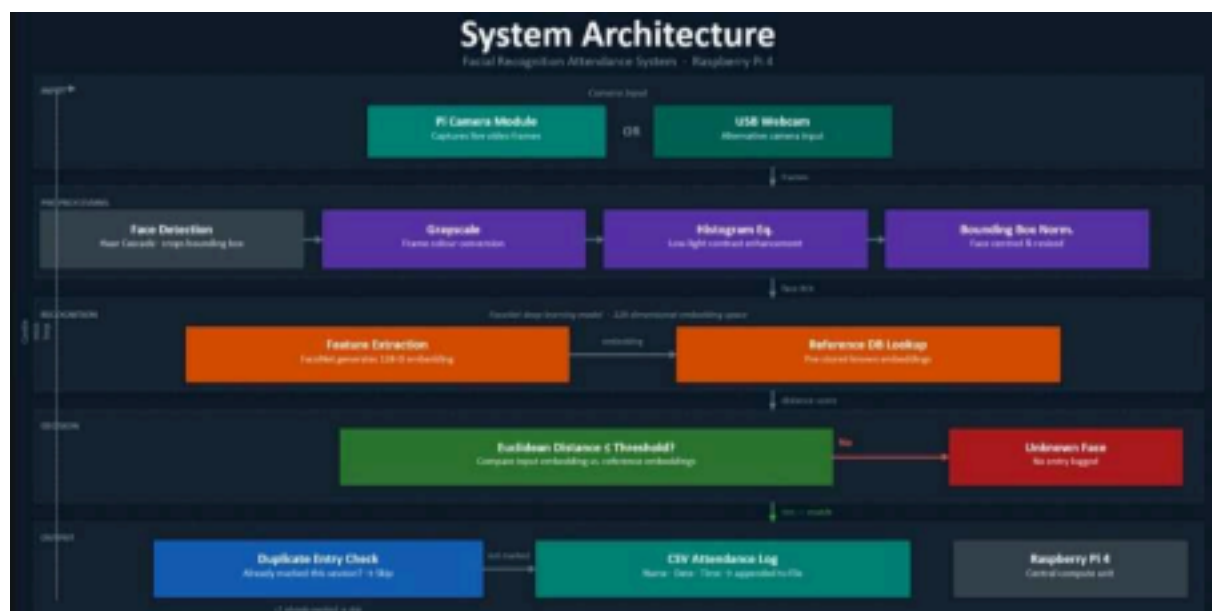
The real-time execution of the system follows the sequence below:

- Initialize camera and load detection and recognition models.
- Load stored embeddings database.
- Capture frame from camera.
- Detect face(s) in the frame.
- Extract and preprocess each detected face.
- Generate embedding using FaceNet.

Compare embedding with database using Euclidean distance.

Apply threshold to determine identity.

- If matched, perform duplicate check.
- Record attendance in CSV file if not already marked.
- Continue processing next frame.



[Figure 3.7: Overall System Architecture]

CHAPTER 4

RESULTS AND ANALYSIS

4.1 Introduction

This chapter discusses the results obtained from the implementation of the facial recognition-based attendance system and analyzes how the system performs in real-time conditions. The evaluation focuses on how accurately the system detects faces, recognizes registered users, handles unknown individuals, and records attendance without errors.

4.2 Experimental Setup

The system was tested using a Raspberry Pi 4 connected to a camera module for capturing live video. A set of registered users was first enrolled into the system by storing their facial embeddings. During testing, multiple individuals were placed in front of the camera, including both registered and unregistered persons.

The testing environment was an indoor setup with normal lighting conditions, similar to a classroom or lab environment. The system was allowed to run continuously to observe its behavior in real-time.

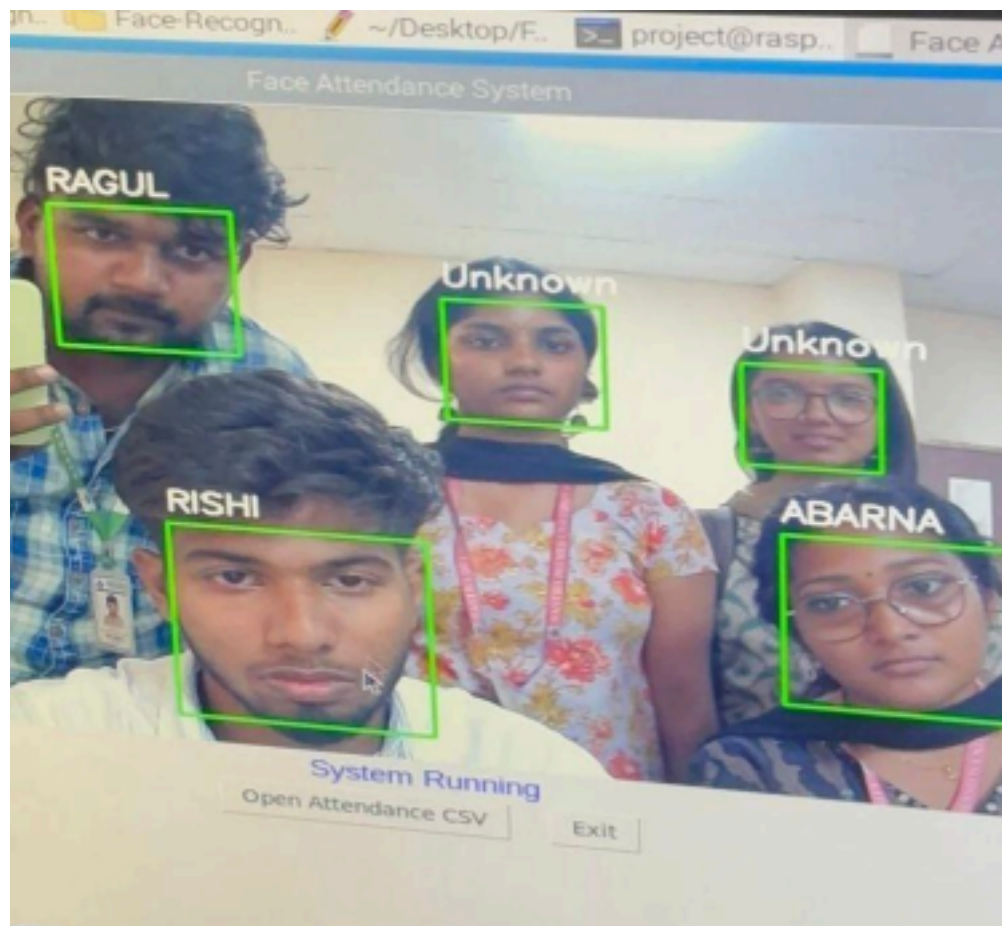
4.3 Real-Time Detection and Recognition

During execution, the system was able to detect faces from the live camera feed and draw bounding boxes around them. Each detected face was then processed for recognition.

From the observed output:

- Trained users such as **RISHI**, **RAGUL** and **ABARNA** were identified correctly.
- Faces that were not present in the database were labeled as “**Unknown.**” ● The system handled multiple faces in a single frame without missing detections.

This confirms that both the detection and recognition stages are functioning properly under real-time conditions.



[Figure 4.1: Facial Recognition]

4.4 Recognition Accuracy

The system uses a distance-based approach to identify individuals by comparing facial embeddings. A threshold value is used to decide whether a face belongs to a registered user or not.

Based on testing:

- Trained individuals were consistently recognized when their faces were clearly visible.
- Unknown individuals were not wrongly classified as registered users.
- Recognition remained stable under normal lighting conditions.

Some variation in accuracy was noticed when:

- The lighting became uneven

- The face angle was slightly turned

Even in such cases, the system mostly maintained correct identification, indicating reliable performance.

4.5 Attendance Recording

After successful recognition, the system records attendance automatically. The output is stored in a CSV file with the name, date, and time.

From the observed results:

- Entries were created immediately after recognition
- The correct timestamp was recorded for each person
- The file format was simple and easy to verify

This confirms that the attendance logging process is directly linked to the recognition step and works without delay.

	A	B	C	D
1	Name	Date	Time	
2	RISHI	20/04/2026	15:37:07	
3	ABARNA	20/04/2026	15:37:07	
4	RAGUL	20/04/2026	15:38:46	
5				
6				
7				
8				
9				

[Figure 4.2: CSV File]

4.6 System Performance

The system was evaluated based on how smoothly it operates in real time.

Observations:

- The camera feed was processed continuously without freezing
- Face detection was quick due to the lightweight classifier
- Recognition and matching were completed within a short time
- Multiple faces were handled without significant delay

Overall, the system performed efficiently on the Raspberry Pi, showing that it is suitable for real-time applications.

4.7 Robustness

The system was tested under different conditions to check how stable it is.

- **Lighting Conditions:**

The system worked well under normal indoor lighting. Slight changes did not affect recognition significantly.

- **Multiple Users:**

It successfully detected and recognized more than one person at the same time.

- **Unknown Faces:**

Faces not present in the database were correctly rejected.

- **Pose Variation:**

Minor changes in face orientation were handled, but extreme angles reduced accuracy slightly.

These observations show that the system is reliable for regular usage environments.

4.8 Limitations

Some limitations were observed during testing:

- Recognition performance decreases in low lighting conditions

- Faces with extreme angles or partial visibility are harder to recognize •

Processing speed is limited by the hardware capability of the Raspberry Pi

These limitations are expected in real-time embedded systems and can be improved with better hardware or optimized models.

4.9 Discussion of Results

The implemented system effectively performed real-time face detection and recognition. It correctly identified registered users, rejected unknown individuals, and recorded attendance without duplication.

The system maintained stable performance during continuous operation and handled multiple users without issues. The results confirm that the proposed approach is practical and suitable for real-world attendance management scenarios.

CHAPTER 5

CONCLUSION

The proposed attendance system demonstrates a practical and efficient way of using facial recognition for automating routine tasks. By combining face detection techniques with a deep learning-based recognition model, the system is able to identify individuals and record attendance without requiring any manual effort. This approach simplifies the overall process and reduces the chances of errors that are commonly seen in traditional methods.

A key aspect of this project is its implementation on a Raspberry Pi platform. This shows that the system is not limited to high-end computing environments and can operate effectively on a compact and low-cost device. The ability to perform real-time detection and recognition on such hardware highlights the feasibility of deploying the system in everyday environments.

The use of FaceNet for generating facial embeddings plays an important role in achieving reliable recognition. Instead of relying on raw images, the system works with numerical representations, which makes comparison faster and more consistent. Supporting steps such as preprocessing and dataset preparation also contribute to maintaining stable performance under normal operating conditions.

Another important outcome of this work is the automation of attendance recording. The system ensures that once a person is recognized, the entry is stored with the correct date and time, while also preventing duplicate entries within the same session. This helps in maintaining organized and accurate records without additional supervision.

In conclusion, the project presents a functional and practical solution for attendance management using modern computer vision techniques. It not only demonstrates the technical feasibility of real-time facial recognition on embedded systems but also shows its relevance in real-world applications. The developed system can serve as a base for further improvements and can be extended with additional features depending on future requirements.

5.1 Future Work

The developed system provides a strong foundation for automated attendance using facial recognition and can be extended in several ways to enhance its functionality and usability. One possible direction is the integration of a centralized database or cloud-based storage.

This would allow attendance records to be accessed and managed remotely, making the system more suitable for large-scale or multi-location deployments.

The system can also be enhanced by developing a user-friendly graphical interface. A dedicated interface would simplify tasks such as user enrollment, attendance monitoring, and report generation, making the system more accessible to non-technical users.

Another important upgrade is the inclusion of multi-camera support. By connecting multiple cameras, the system can cover larger areas and monitor multiple entry points simultaneously, which is useful in bigger classrooms or organizational setups.

Further improvements can involve optimizing the recognition process to handle a larger number of users efficiently. This would allow the system to scale without affecting real-time performance, making it more adaptable to institutions with higher attendance volumes.

The system can also be integrated with existing institutional platforms, such as student information systems or employee management software. This would enable automatic updating of attendance records and reduce the need for manual data handling.

Additionally, Future versions may also explore mobile or web-based access, allowing users to view attendance reports and system status from remote devices.

Overall, these enhancements can transform the current implementation into a more scalable, user-friendly, and fully integrated attendance management solution suitable for modern smart environments.



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Signature of the guide:

CHAPTER 6

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